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OCR Hinders RAG: Evaluating the Cascading Impact of OCR on Retrieval-Augmented Generation

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Abstract

Retrieval-augmented Generation (RAG) enhances Large Language Models (LLMs) by integrating external knowledge to reduce hallucinations and incorporate up-to-date information without retraining. As an essential part of RAG, external knowledge bases are commonly built by extracting structured data from unstructured PDF documents using Optical Character Recognition (OCR). However, given the imperfect prediction of OCR and the inherent non-uniform representation of structured data, knowledge bases inevitably contain various OCR noises. In this paper, we introduce OHRBench, the first benchmark for understanding the cascading impact of OCR on RAG systems. OHRBench includes 8,561 carefully selected unstructured document images from seven real-world RAG application domains, along with 8,498 Q&A pairs derived from multimodal elements in documents, challenging existing OCR solutions used for RAG. To better understand OCR's impact on RAG systems, we identify two primary types of OCR noise: Semantic Noise and Formatting Noise and apply perturbation to generate a set of structured data with varying degrees of each OCR noise. Using OHRBench, we first conduct a comprehensive evaluation of current OCR solutions and reveal that none is competent for constructing high-quality knowledge bases for RAG systems. We then systematically evaluate the impact of these two noise types and demonstrate the trend relationship between the degree of OCR noise and RAG performance. Our OHRBench, including PDF documents, Q&As, and the ground truth structured data are released at: https://github.com/opendatalab/OHR-Bench

1. Introduction

Retrieval Augmented Generation (RAG) enhances Large Language Models (LLMs) by integrating external knowledge [17, 25], enabling them to respond accurately to queries beyond their training corpus, such as recent news or proprietary content, and reducing hallucinations [23, 25, 38, 43]. This is achieved through a retrieval-then-grounding approach, where relevant documents are retrieved from external knowledge bases and incorporated into the LLM's prompt for grounding.

As an essential component of RAG systems, the knowledge base defines the scope and quality of documents that RAG can access. Given that a vast amount of real-world knowledge resides in unstructured documents, such as scanned PDFs, constructing an external knowledge base often relies on Optical Character Recognition (OCR) [1] to parse structured data from these unstructured PDF documents [20, 57]. For instance, MinerU [48] takes raw PDFs as input and extracts plain text, formulas, and tables into structured formats for subsequent RAG applications. However, imperfect predictions of OCR and non-uniform representations of parsing results impair the construction of a high-quality knowledge base for RAG. To be specific, despite advancements in OCR [5, 48, 50], even the leading model cannot achieve perfect accuracy across all scenarios [31, 57]. Furthermore, structural data like table can inherently be parsed in different representation, such as Markdown or LaTeX. These issues introduce OCR noise in parsing results and diminish the quality of the knowledge base. Considering RAG is sensitive to input noise [11, 15, 54], recent works race on downstream RAG components, including more precise retrievers [7, 26, 34] and more advanced LLMs [1, 13, 15, 55]. However, the quality of OCR-based external knowledge bases and its cascading impact on these downstream RAG components have received less attention, which highlights a critical but unaddressed gap: *the absence of benchmarks to assess OCR's cascading impact on each component and entire system of RAG.*

Existing benchmarks either evaluate RAG holistically

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[1]We employ the General OCR concept for document parsing from GOT-OCR2.0 [50], which includes, text recognition, multimodal data extraction (table, formula, and chart recognition), and reading order restoration.

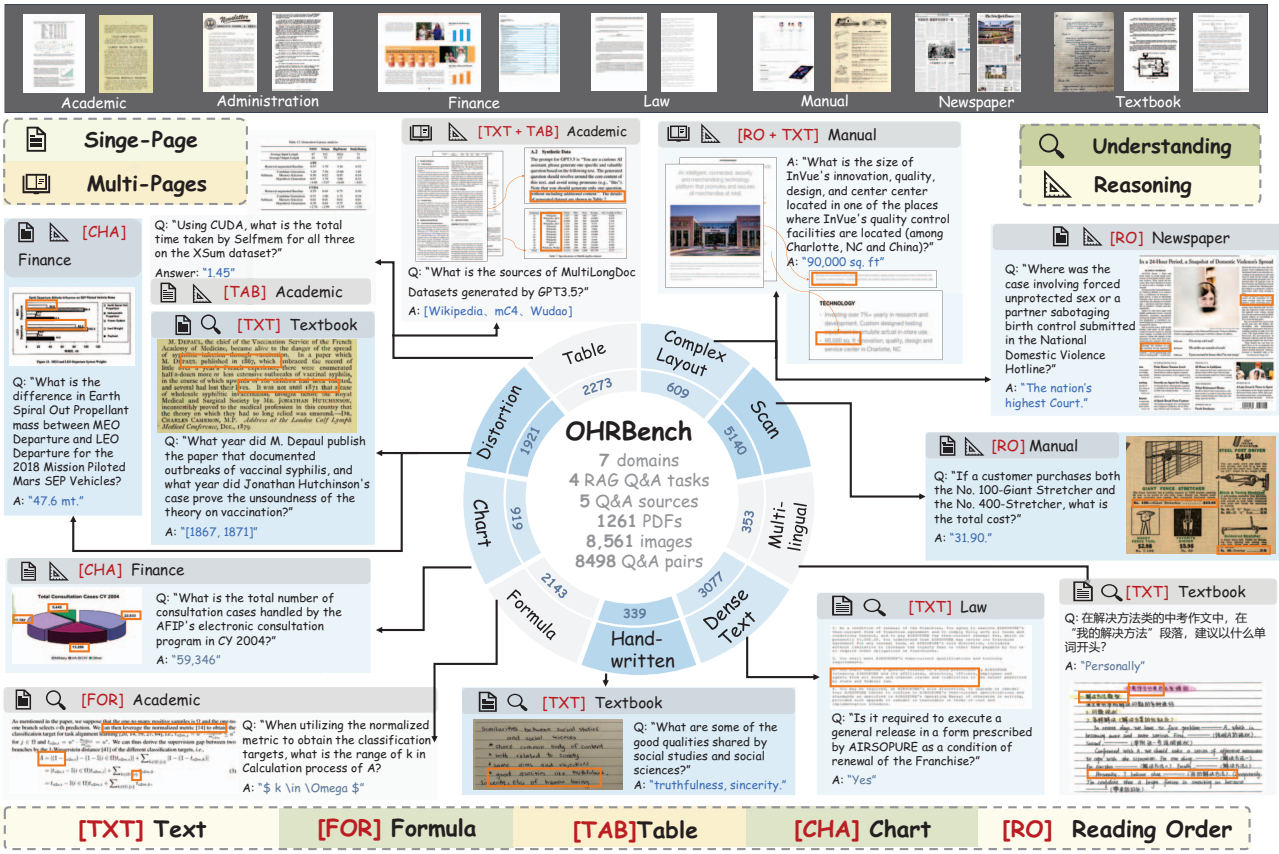


Figure 1. Our OHRBench comprises documents from 7 domains, 9 challenging attributes for OCR, 4 types of Q&A tasks, and 5 Q&A evidence sources. Each number indicates the count of PDF pages with that attribute. Criteria for these attributes can be found in Appendix Sec. II.3

without fine-grained assessment [58], consider limited OCR solutions without accounting for the noise they introduce [16, 20]. Additionally, they lack documents that present more diverse OCR challenges, such as scanned historical, multilingual, and handwritten documents. To fill this gap, we introduce OHRBench, a question-answering benchmark designed to evaluate OCR's cascading impact on each component and entire systems of RAG in two ways. First, we construct a document-based RAG Q&A dataset comprising complex, unstructured PDF documents from 7 RAG real-world application areas: *Textbook, Law, Finance, Newspaper, Manual, Academic and Administration*. As detailed in Tab. 1 and Fig. 1, we have collected 8,561 document images featuring attributes that challenges the creation of high-quality knowledge bases for RAG systems. We also provide diverse Q&A pairs which not only span realistic RAG tasks, including understanding, reasoning, and multi-page questions, but also features evidence sourced from key components of OCR in document parsing, making them ideal for assessing the OCR's impact on RAG performance. Second, we identify two primary OCR noise types: *Semantic Noise*, resulting from prediction errors, and *Formatting Noise*, arising from diverse document element representation. By systematically introducing these noise types into documents, we generate perturbed structured data with varying degrees of noise, enabling further exploration of the quantitative relationship between OCR noise and RAG performance.

With OHRBench, we first conduct a comprehensive benchmark on current OCR solutions, including pipeline-based OCR systems [36, 48], end-to-end OCR models [5, 50] and Vision-Language Models (VLMs) for OCR [4, 8, 9, 49, 51, 52]. We reveal that even the best OCR solutions exhibit a performance gap of 14% at least, compared to the ground truth structured data, facilitating the importance of mitigating OCR noise in RAG systems. Further experiments on different types of OCR noise uncover that *Semantic Noise* consistently exert a significant impact, while *Formatting Noise* affects specific retrievers and LLMs differently, offering valuable insights for developing RAG-tailored OCR solutions and noise robust models.

Contributions. We summarize our main contributions:

- We present OHRBench, a question-answering benchmark designed to evaluate the impact of OCR on RAG systems. OHRBench includes various unstructured PDF documents from seven RAG domains with ground truth structure data annotations and Q&A pairs spanning multiple RAG tasks

with diverse source of evidences, posing challenges to the employment of current OCR solutions in RAG systems.
- We conduct a comprehensive evaluation of current OCR solutions and reveal that none of them is competent for constructing high-quality knowledge bases for RAG systems.
- We identify two primary types of OCR noise, including *Semantic Noise* and *Formatting Noise*, generate perturbed data with varying levels of noise and explore the trend relationship between the degree of OCR noise and RAG performance.

2. Related Works

2.1. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [23, 25, 38] integrates external knowledge into large language models (LLMs) to mitigate hallucinations. Although RAG technology enhances the generation capabilities of LLMs, it is notably sensitive to input noise. InfoRAG [54] characterizes this noise in RAG as incorrect and irrelevant content within retrieved text and reveals its impact on RAG performance. RAAT [15] further expands noise into relevant noise, counterfactual noise, and irrelevant types. However, these studies focus solely on chunk-level noise introduced during the retrieval stage and its effect on the generation capabilities of LLMs, leaving the impact of noise derived from OCR results unexplored. GARAG [11] examines typographical errors, a form of OCR noise, but its scope is limited to plain text using only synthetic data, overlooking the variety of OCR noise encountered in real-world RAG applications. In this paper, we reveal the impact of noise introduced during the OCR stage, offering a comprehensive analysis of its impact on RAG systems.

2.2. Document parsing with OCR

OCR-based document parsing is a promising solution for structured data extraction from unstructured documents, facilitating applications [30, 32, 57] like RAG. Current OCR solutions can be summarized into three categories, pipeline-based systems [36, 47, 48], end-to-end models [5, 29, 50], and employing VLMs for OCR [2, 3, 9, 21, 49] Pipeline-based systems decompose OCR into multiple subtasks, such as layout detection, text, formula, and table recognition, enabling fine-grained data extraction. End-to-end models take document images as input and output the overall recognition result in an end-to-end manner. Due to the achievement of VLMs on visual understanding, recent works have explored its application in OCR [31, 51]. In this paper, we evaluate these OCR paradigms, examining their suitability for RAG applications across diverse, real-world document domains.

2.3. Benchmark and evaluation of Retrieval-Augmented Generation

Frameworks like RAGAS [14] and ARES [40] propose evaluating RAG systems based on context relevance, answer faithfulness, and answer relevance, using LLMs or fine-tuned discriminators for measurement. RGB [6] assesses the noise robustness, negative rejection, information integration, and counterfactual robustness of RAG in news data. MultiHop-RAG [42] focuses on multi-hop reasoning capabilities, while ClashEval [53] explores the context preference in conflicting evidence scenarios. However, these evaluations target specific components of RAG systems, and none of them discusses the impact of external knowledge base construction on RAG systems. Although UDA [20], VisRAG [56] and M3DocRAG [10] explore RAG's effectiveness in document understanding, they consider limited OCR solutions, ignoring challenging documents and lacking analysis of different OCR noise types. In this paper, we introduce OHRBench to comprehensively investigate OCR noise's impact on RAG systems.

3. OHRBench

Our OHRBench consists of 1) a number of unstructured PDF documents from seven real-world RAG applications, Q&A pairs derived from multimodal document elements and ground truth structure data annotation, and 2) perturbed structured data based on ground truth with varying degrees of OCR noises. Fig. 2 illustrates the construction of OHRBench. We will now delve into the details of each component.

3.1. Data collection

According to [27, 57], extracting structured data from multimodal document elements like formulas and tables poses significant challenges to current OCR solutions. Considering the practical application scenarios of RAG and the challenging field of OCR, we compile a PDF document collection representing seven common RAG application scenarios: *Textbook, Law, Finance, Newspaper, Manual, Academic and Administration*. This collection includes a diverse array of documents from both existing datasets and public web resources. Specifically, we first collect PDF documents from a wide range of existing datasets, including DUDE [46], OmniDocBench [35], FinanceBench [22], CUAD [18], and GNHK [24]. This results in a highly diverse PDF dataset that encompasses complex structured data and layouts, high text density, handwritten, scanned, and historical documents, as well as multilingual content (Chinese and English), which covers most challenges faced by OCR in document parsing. In addition to existing datasets, we further supplement our collection with documents from public resources to balance the distribution. We filter out the

Statistic	Number
Documents	1,261
- Domain	7
- Total Pages	8,561
- Avg.Tokens	1,034/page
- Avg.Data Type	1.9/page
Questions	8,498
Avg.Question Token	17.9 ± 8.7
Avg.Answer Token	3.5 ± 4.1
(Evidence Source)	
- Text	3,528 (41.5%)
- Table	2,364 (27.8%)
- Formula	1,267 (14.9%)
- Chart	768 (9.0%)
- Reading Order	691 (8.1%)
(Answer Format)	
- String	4,171 (54.4%)
- Numeric	3,004 (35.3%)
- Yes/No	594 (7.0%)
- List	483 (6.9%)
(Task Type)	
- Understanding	6,114 (71.9%)
- Reasoning	2,384 (28.1%)
- Single-page	7,656 (90.1%)
- Multi-page	842 (9.9%)

Table 1. Dataset Statistics

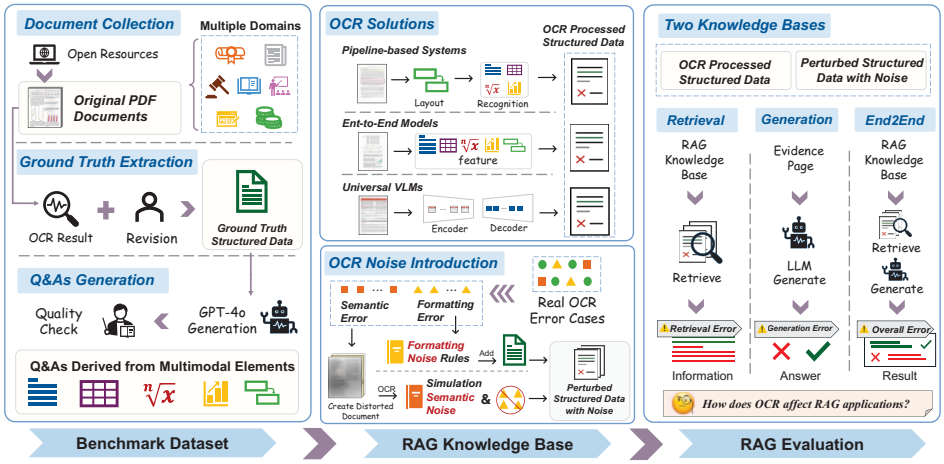


Figure 2. Construction of OHRBench and evaluation protocol. (1) Benchmark Dataset: documents from seven domains, human-verified ground truth structured data, and Q&As from multimodal document elements. (2) RAG Knowledge Base: Current OCR results for benchmarking and perturbed data for assessment. (3) Evaluation of OCR impact on each component and the overall RAG system.

corrupted or license-restricted documents and finally curate a document dataset comprising 1,261 PDFs and 8,561 images. For each collected document, we manually categorize them into 7 domains and provide ground truth structured data. Specifically, we begin with parsing all documents using Mathpix[2] for structural data extraction. We then ask expert-level annotators to revise the results, ensuring fidelity to the original structure and content of PDFs while mitigating any style deviations from Mapthix. Detailed descriptions of our selection and processing pipeline can be found in the Appendix Sec. II.

3.2. Q&A pairs generation

The process of extracting structured data from documents involves three key tasks: recognizing plain text; extracting multimodal document elements, including tables, formulas, and charts; and restoring reading order which includes multi-column and truncated paragraph merging. To systematically assess the impact of OCR results on RAG performance, our Q&A generation approach revolves around these 5 evidence sources and various realistic Q&A tasks. Specifically, we provide the ground truth structured data of each document page to GPT-4o and prompt it to generate Q&A based on important components in document parsing, including plain text, tables, formulas, and charts. For questions related to reading order, we identify paragraphs that require merging and instruct GPT-4o to create questions that necessitate combining these paragraphs for a complete answer. We generate both understanding questions, which only require extracting specific information, and reasoning questions, which involve arithmetic operations, comparisons, or synthesizing information across multiple sections. For multi-page Q&A, we derive them from both single-page Q&A and ground truth structured data from different pages that share the same entity name, recognized with spaCy [19]. Detailed process and the prompt template for the Q&A generation are provided in Appendix Sec. I. Each Q&A consists of the following fields: one page of the original PDF document, evidence context from ground truth structured data that provides the answer to the question, type of evidence (plain text, table, formula, chart and reading order), and the question and answer which are both derived from this evidence context. In this way, these Q&As can serve as a testbed for evaluating OCR results on multimodal document elements.

Quality Control. The quality of Q&A pairs generated by a large language model (LLM) can vary significantly. To address this issue, we apply three data selection criteria to ensure high-quality Q&As: (1) compatibility with realistic RAG applications, (2) faithfulness to task definitions and (3) correctness. We incorporate both heuristic methods and prompting LLM for auto data filtering:

- **Compatibility to RAG Applications.** Questions should be context-independent and not answerable by the model's internal knowledge. We collect keywords from existing context-dependent questions, such as "according to the document", as heuristic rules. Following [56], we also employ LLMs to classify context dependence for further filtering. Questions answerable without retrieval are excluded by instructing LLMs to answer without access to the evidence context.
- **Faithfulness to Task Definition.** We ensure questions align with their task definition (reasoning or understanding) using LLMs to judge and that evidence sources match the context using heuristic rules. In addition, for multi-page question, we provide single evidence context and use LLMs' responses to filter out answerable questions.

[2] https://mathpix.com/

- **Correctness.** We verify the accuracy of both evidence context and answers. In specific, we provide LLMs with oracle evidence contexts and sample answers repeatedly to filter Q&A below a certain correctness threshold.

Finally, we manually check against the criteria to ensure the quality. The LLMs used in our verification include GPT-4o and DeepSeek-V3 [28], where we find that DeepSeek-V3 achieve similar performance compared to GPT-4o on Q&A verification. A detailed description can be found in the Appendix Sec. II.5. This multi-step quality control ensures the Q&A dataset meets diverse evidence source requirements and practical RAG applications. Ultimately, we filter out 8498 high-quality Q&As from 15317 candidates.

3.3. Data perturbation with OCR noise

Despite advancements in OCR, real-world applications often encounter document types beyond the training corpus of OCR models, leading to low-quality data extraction. Additionally, the different structured representations of document elements further introduces noise, impacting RAG performance. In this paper, we focus on two key types of OCR noises: ***Semantic Noise*** and ***Formatting Noise***. To quantitatively analyze their effects, we start from errors in current OCR results and generate perturbed data with different noise levels. We will now delve into the details of each type.

Semantic Noise results from OCR prediction errors that impact the semantics of parsed content, deviating retrievers and LLMs from integrating correct information related to user queries. To systematically capture realistic *Semantic Noise*, we include diverse perturbation to document images and utilize multiple OCR solutions to perform OCR on these document images, capturing a wide range of real-world *Semantic Noise* as much as possible. We begin with collecting naturally distorted documents and identifying common degradation patterns, such as background artifacts, watermarks, and structural distortions (e.g., dilation and erosion). Then, we extend from [5], where its method has been shown to be effective for simulating naturally distorted documents, we refine perturbation strategies through an iterative, cross-validated process involving multiple annotators. One annotator adjusts distortion parameters and applies them to document images, while a another annotator, who is unaware of the applied modifications, distinguish which document appears artificially altered. This refinement continues until the perturbations become indistinguishable from real samples. Through this process, we identify 8 effective perturbation types that balance realism and intensity. Details and examples are provided in Appendix Sec. III.2. By varying the number and type of perturbations, we generate 3 distinct datasets with controlled *Semantic Noise* levels. We then choose MinerU, GOT, and Qwen2.5-VL to curate 9 perturbed data with diverse appearance of *Semantic Noise*, enabling a systematic evaluation of its impact on RAG performance.

Formatting Noise stems from stylistic commands, such as white space characters for beautifying formulas and bold and italic commands for better readability, and inconsistencies in structured data representations across Markdown, LaTeX, and HTML. Although irrelevant to semantics, this noise complicates information integration for both retrievers and LLMs. To assess the impact of *Formatting Noise* on RAG, we identify common OCR-induced formatting inconsistencies and develop heuristic rules to introduce controlled perturbations through additions, removals, and format conversions. A detailed list of perturbation rules is in Appendix Sec. III.1. By applying these modifications at varying proportions in ground truth structured data, we create three datasets with different degrees of *Formatting Noise*. Additionally, we evaluate RAG performance under different structured data formats, comparing retrieval and reasoning consistency across Markdown, LaTeX, and HTML representations.

4. Experiments

4.1. Experimental settings

We evaluate the impact of OCR on RAG systems in three ways: retrieval performance, generation performance, and overall system performance. For the retrieval stage, we utilize knowledge bases derived from the same domains of user queries and retrieve the top-2 matched chunk. During the generation stage, we provide the page where the question is derived from for LLMs to generate the response. In the overall evaluation, retrievers retrieve the relevant chunks from the knowledge base in the same domain as the question, and LLMs generate responses based on these chunks. In the overall evaluation, we provide the top-2 matched chunk for generation unless otherwise stated. The default chunk size is 1024 with no overlap.

Metrics. To evaluate the quality of OCR results, we calculate the edit distance between each page of OCR results and the ground truth structured data and report the average values. For assessing retrieval performance, as results of different OCRs often include various extraneous characters, discriminating whether the evidence exactly appears in the retrieved contents is not fair. Following [20], we employ Longest Common Subsequence (LCS) [37] to measure evidence inclusion in retrieved content. For the generation stage, we employ the F1-score metric to measure the accuracy of LLMs' responses.

Retrievers. We consider two primary retrievers: (1) BGE-M3 [7], a recent SOTA dense retriever within its size category. (2) BM-25 [39, 45] is a lightweight sparse retriever ranking document based on the query term frequency.

LLMs. We employ three representative open-source LLMs: Qwen2 (Qwen2-7B-Instruct and Qwen2-72B-Instruct) [49] and Llama-3.1 (Llama3.1-8B-Instruct) [13]. A standard

	OCR	Retrieval						Generation						Overall					
	E.D.↓	TXT↑	TAB↑	FOR↑	CHA↑	RO↑	ALL↑	TXT↑	TAB↑	FOR↑	CHA↑	RO↑	ALL↑	TXT↑	TAB↑	FOR↑	CHA↑	RO↑	ALL↑
Ground Truth	-	81.2	69.6	74.8	70.3	9.8	70.0	49.4	46.0	34.0	47.0	28.2	43.9	45.0	34.6	28.0	32.9	18.7	36.1
Pipeline-based OCR																			
MinerU [48]	0.24	67.7	48.5	51.1	16.5	**5.9**	50.1	**45.9**	39.3	28.6	9.7	**29.5**	36.7	**41.4**	28.5	23.0	9.3	**17.8**	30.0
Marker [36]	0.28	**75.2**	57.8	55.4	19.7	**5.9**	56.6	44.5	37.8	27.8	10.9	26.2	35.9	40.1	28.1	22.3	10.0	16.2	29.5
End-to-end OCR																			
GOT [50]	0.27	62.1	41.0	48.7	17.4	3.7	45.4	37.5	28.5	24.1	8.5	7.1	27.8	35.3	22.9	20.1	8.2	5.3	24.6
Nougat [5]	0.34	59.1	32.7	44.2	11.3	4.4	40.9	36.7	22.9	22.9	6.4	6.9	25.5	33.5	18.4	19.4	5.8	3.6	14.5
Vision-Language Model for OCR																			
Qwen2.5-VL-72B [4]	0.18	74.6	**59.8**	**59.7**	38.2	5.3	**59.2**	44.4	**42.1**	**31.8**	**27.0**	11.6	**37.5**	40.6	**31.1**	**26.1**	19.0	8.8	**31.1**
InternVL2.5-78B [8]	0.28	68.2	57.7	55.3	**45.1**	2.7	**55.8**	41.8	41.8	29.0	**33.6**	3.3	35.8	38.2	31.0	23.3	**22.9**	3.1	29.6

Table 2. Evaluation of various OCR solutions and their impacts on RAG systems. The OCR performance is reported using edit distance (E.D.). We report the generalized LCS or F1 of five types of evidence sources, including plain text (TXT), table (TAB), formula (FOR), chart (CHA), and reading order (RO). **Bold** indicates the best performance, and underline indicates the second-best performance.

prompt template is used to format responses consistently across all LLMs (see Appendix Sec. I). All open-source models are downloaded from Huggingface [3], with inference conducted on 8 NVIDIA A100 GPUs.

4.2. Benchmarking current OCR solutions

In this section, we evaluate the suitability of current OCR solutions for real-world RAG applications by conducting comprehensive experiments with our OHRBench. We involve several representative OCR solutions including (1) Pipeline-based OCR, such as MinerU [48] and Marker [36], (2) End-to-end OCR, including GOT [50] and Nougat [5], and (3) Vision-Language Models, specifically Qwen2.5-VL-72B [44] and InternVL2.5-78B [8]. For GOT, we employ its format OCR mode to output structured data. For Qwen2.5-VL-72B and InternVL2.5-78B, we prompt them to produce formulas, tables, and charts in LaTeX format, with the prompt template available in Appendix Sec. I. The retrievers used are BGE-M3 and BM25, while the LLMs are LLama-3.1-8B-Instruct and Qwen2-7B-Instruct. All metrics are averaged across domains and combinations of retrievers and LLMs. Details of experimental results are available in Appendix Sec. IV.5.

Through the comparison presented in Tab. 2, we derive several key conclusions about the performance of these OCR solutions and their corresponding impacts on RAG systems, as follows: **(1) VLMs for OCR achieve the best overall performance.** Among all OCR solutions, Qwen2.5-VL-72B consistently outperforms others across all three evaluation stages. Its superiority stems from its ability to handle structured data more effectively than both pipeline-based and end-to-end OCR methods. Despite claims that GOT can parse charts, its performance remains subpar. Similarly, MinerU, Marker, and Nougat fail to produce comparable results due to their inability in parsing chart. For plain text questions, although its poor performance on high-resolution documents (see newspaper in APPendix Sec. IV.5, Qwen2.5-VL-72B performs comparably to pipeline-based OCR. Our manual review suggests that its strong language decoder enhances robustness against historical and distorted documents, a capability lacking in pipeline-based OCR. **(2) Reading order is challenging for VLMs and End-to-end OCR.** Despite their strong semantic understanding capabilities offered by language decoders, both VLM and end-to-end OCR struggle with merging paragraphs correctly, reflected in F1-score of just 8.8 in overall evaluation. In contrast, pipeline-based OCR, though lacking semantic understanding, achieves performance close to ground truth (GT) using rule-based strategies. However, GT itself performs poorly, likely because reading-order questions require integrating information across multiple paragraphs, posing challenges for current RAG systems [42]. **(3) All OCR solutions exhibit performance degradation.** Even the best solutions experience a 14% (5 F1-score) drop in the overall stage evaluation, with greater losses in the retrieval and generation stages. This indicates that our OHRBench presents significant challenges for both OCR solutions and RAG systems.

In summary, current OCR solutions struggle to maintain robustness and effectiveness across diverse real-world RAG application scenarios. Additionally, standard OCR metrics like edit distance do not always align with RAG performance. For example, while MinerU and Qwen2.5-VL-72B exhibit lower edit distances compared to Marker, they do not consistently achieve better performance across all metrics. This discrepancy may be attributed to the varying types of OCR noise introduced by different solutions. To further investigate, we systematically explore the impact of these OCR noise types on RAG in Sec. 4.3.

4.3. In-depth analysis of OCR noise's impact on RAG

In this section, we conduct an in-depth analysis of the impact of *Semantic Noise* and *Formatting Noise* on RAG systems, using perturbed structured data with varying levels of perturbations. For each type of OCR noise, we introduce three noise levels—mild, moderate, and severe—to systematically assess their effects. Since our findings in Sec. 4.2 indicate that edit distance fails to accurately capture the degree of

[3] https://huggingface.co/

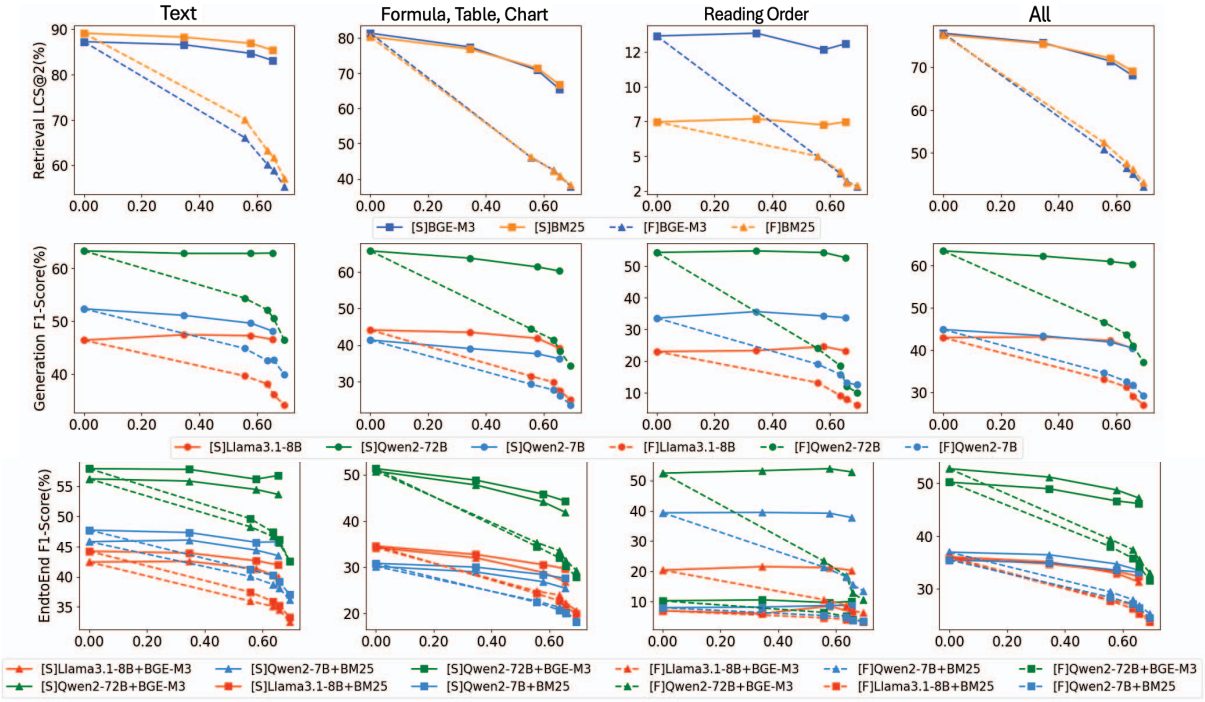


Figure 3. Impact of *Semantic Noise* ([S] dashed lines) and *Formatting Noise* ([F] solid lines) on RAG components. The horizontal axis denotes the ratio r_{noise}, where higher values indicate greater OCR-induced noise. We report LCS and F1-score for each evidence source: text (first column), the average score for multimodal elements (tables, formulas, and charts, second column), reading order (third column), and all sources combined (last column).

OCR noise, we instead define the ratio r_{noise}, representing the proportion of Q&A pairs affected by OCR noise as a measure. Specifically, for each Q&A pair, we compute the LCS between its evidence context and the corresponding perturbed structured data. If LCS exceeds 0.95, the Q&A pair is considered as unaffected, otherwise, it is affected. We then use the ratio (r_{noise}) to quantify the degree of perturbations, with 0 representing to ground truth structured data and values approaching 1 indicating greater perturbation. This approach allows us to align the perturbation levels of *Formatting Noise* and *Semantic noise* for fair comparisons. Additionally, for *Formatting Noise*, we evaluate its retrieval performance with modified LCS calculations by excluding stylistic commanding introduced during perturbation, ensuring a fair assessment of retrieval accuracy. For each degree of *Semantic Noise*, we report the average RAG performance using three different OCR results, including MinerU, GOT, and Qwen2.5-VL-72B.

4.3.1. Fine-grained impact on retrieval and generation

***Semantic Noise* significantly influences both retrieval and generation phases.** As illustrated in Fig. 3, increasing *Semantic Noise* from mild ($r_{\text{noise}} = 0$) to severe ($r_{\text{noise}} > 0.6$) results in nearly a 50% performance decline for most retrievers and LLMs. In the retrieval stage, both the sparse retriever BM25 and the dense retriever BGE-M3 suffer consistent performance declines across all types of questions, suggesting that dense retrieval's stronger comprehension does not provide robustness against *Semantic Noise*. In the generation phase, all LLMs struggle with *Semantic Noise*, among which performance on reading-order decreases the most. Interestingly, although the way we introduce *Semantic Noise* should primarily affect text recognition, questions related to multimodal elements (tables, formulas and charts) degrade even further, highlighting the challenges in parsing, understanding and reasoning over multimodal document data.

***Formatting Noise* primarily affects multimodal questions.** While performance on plain text queries and reading-order-related questions remain largely unaffected, retrieval and generation performance drops more severe for multimodal queries. The maximum performance losses reach 12.7% for BGE-M3 and 9.1% for Llama3.1-8B in retrieval and generation, respectively. In addition, larger LLMs exhibit greater robustness, with only a 7% performance reduction on multimodal questions, indicating that more advanced LLMs can effectively handle *Formatting Noise*.

4.3.2. Impact on end-to-end evaluation

***Semantic Noise* consistently demonstrate a strong impact, while *Formatting Noise* affects specific retrievers and LLMs differently.** *Semantic Noise* consistently degrades performance across all combination of retrievers and LLMs, particularly on multimodal questions involving tables, formulas, and charts. In contrast, the effect of *Formatting Noise* is more variable when using smaller LLMs, Llama3.1-8B and Qwen2-7B, for generation, despite greatly reduced retrieval

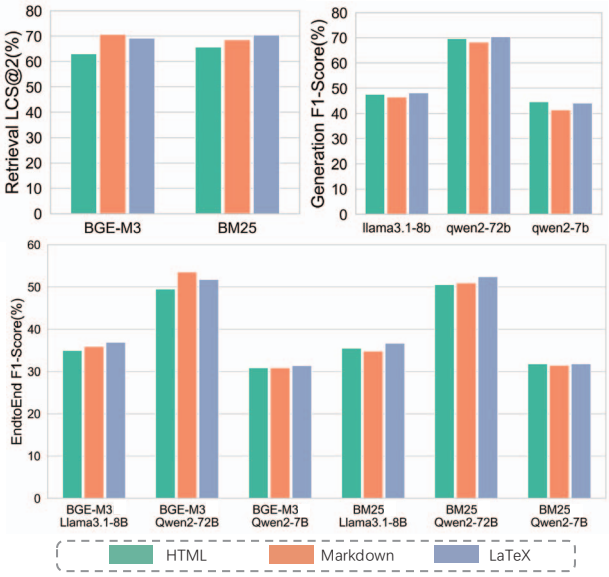


Figure 4. Performance of retrieval, generation and end-to-end with different table format. We only report the results of table-related questions.

accuracy, the overall performance shows a slight change due to their limited information integration capabilities. Conversely, using a larger LLM, Qwen2-72B, is highly sensitive to retrieval performance. While its generation performance shows light changes, the overall performance decreases, especially on questions related to multimodal elements.

In summary, *Semantic Noise* significantly affects each stage of RAG and the entire system. The impact of *Formatting Noise* varies with different retrievers and generators, particularly affecting questions with multimodal elements.

4.3.3. Impact of table format

In addition to perturbations, we investigate the influence of different table formats as a kind of *Formatting Noise*. As shown in Fig. 4, HTML tables show inferior performance during retrieval compared to the Markdown and LaTeX formats. Markdown and LaTeX formats perform similarly, with BGE-M3 demonstrating a better understanding of Markdown. In the generation phase, HTML and LaTeX showed similar performance across all models, but the Markdown format performed worse due to the lack of support for merging cells. In end-to-end evaluations affected by low retrieval performance, using HTML tables is comparatively worse, while the combination of Qwen2-72B and BGE-M3 using Markdown achieves the best performance.

4.3.4. Error analysis

We further conduct error analysis to understand the bottlenecks of the RAG system with OCR noise in a quality approach. Specifically, we calculate the distribution of OCR errors and RAG errors when using ground truth structured data and perturbed data with severe *Formatting Noise* and

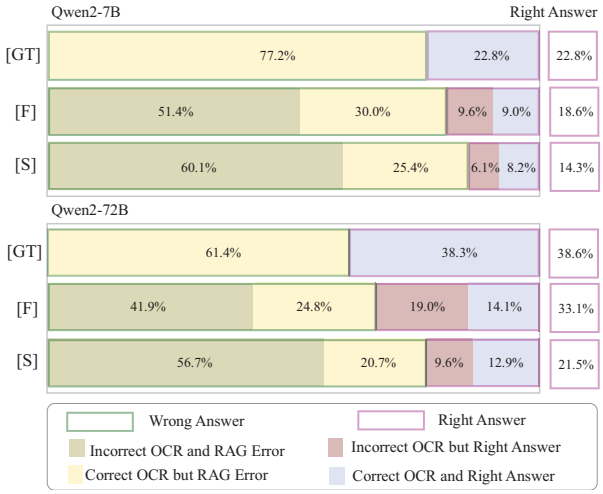


Figure 5. Analysis of answer correctness distribution in Q&A pairs, using different perturbed data and LLMs. It reveals that larger LLMs are more robust to OCR noise. [S] and [F] denote perturbed data with severe *Semantic Noise* and *Formatting Noise*, respectively. [GT] represents ground truth perturbed data.

Semantic Noise. Our evaluation employs BGE-M3 as the retriever and assesses error distributions using two generators: Qwen2-7B and Qwen2-72B. We use the same strategies to identify the proportion of OCR errors as in Sec. 4.3. The distribution of these errors is illustrated in Fig. 5. It indicates that when the proportion of OCR errors is similar for *Semantic Noise* and *Formatting Noise* (66% vs. 61%), it performs better with *Formatting Noise*. Of these, about half of the correct responses when using Qwen2-7B as a generator were done by the model with incorrect OCR results. Meanwhile, larger LLMs are more robust to OCR noise. Compared to Qwen2-7B, Qwen-72B has a 9.4% and 3.5% higher percentage of samples with OCR errors but ultimately correct in *Formatting Noise* and *Semantic Noise*, respectively.

5. Conclusion

In this paper, we present OHRBench to evaluate the impact of OCR on RAG systems, which encompasses diverse PDF documents from sven RAG application scenarios along with Q&A pairs derived from multimodal elements in these documents. Through comprehensive evaluations of current OCR solutions, we reveal that none is fully capable of RAG systems across all scenarios. Furthermore, our analysis of different types of OCR noise demonstrates that while no retrievers and LLMs are immune to *Semantic Noise*, more advanced models exhibit greater resilience to *Formatting Noise*. We believe that our documents featuring challenging OCR attributes and Q&A pairs sourced from varied document elements, will advance the development of OCR solutions tailored for RAG and OCR noise-resistant RAG systems.

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